School of Future Tech

Case Study Report

On

Employee Payroll System

by

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1. Introduction to Case study

- Employee Payroll System is a fundamental application used in organisations to manage and calculate employee salaries efficiently. It automate calculation of allowances, deductions, reducing manual errors and saving time.
- This case study focus on developing a simple payroll management program using C++**.** The program allows the user to input employee information, perform salary calculations based on predefined rules, and generate a payslip displaying salary details.
- Objective of this case study is to apply Object-Oriented Programming (OOP) concepts such as classes, objects, and functions while also understanding structured programming and salary computation logic.

2. Problem Statement/Case Background

Background

Payroll management is an important task in every organization as it ensures employees receive their salaries correctly and on time. Manual payroll systems can lead to calculation errors and inefficiency. Therefore, automated payroll systems are used to simplify salary calculations and record management. This case study focuses on developing a simple **Employee Payroll System using C++** to manage employee salary information and generate payslips.

Abstract

Employee Payroll System is a C++ console-based application that allows users to input employee details such as Employee ID, Name, and Basic Salary. The program calculates salary components like allowances and generates a payslip. It uses **classes, functions, and a menu-driven approach** to perform payroll operations efficiently.

3. Problem Statement/Case Study Design

   This is designed using Object Oriented Programming principles. MainComponent of program is Employee class, which stores employee details

   The program contains following main features:

   1. Employee Information
      - Employee ID
      - Employee Name
      - Basic salary
   2. Payroll Calculation
      - Calculates allowances such as:
      - House Rent allowance(HRA)
      - Dearness allowance(DA)
      - Bonus
      - Calculate Gross Salary
   3. Payslip Generation

      Display employee salary details in structured format.
   4. Menu-driven Control System

      Allows user to choose options
      - Add Employee
      - Generate Payslip
      - Exit Program

4. Methods and Technology applied in the Problem statement/Case Study

Case study used several programming concepts

1. Object-Oriented Programming(OOP)

System is designed using class named Employee which contains:

- Data members to store employee details.
- Member functions to perform operations.

2. Menu-driven Program

It uses a loop and switch statement to create a menu-driven system that allows users to interact with the program again and again until they choose to exit.

Example logic:

Display menu options

Accept user choice

Execute corresponding function

3. Salary Calculation Algorithm

Salary is calculate using simple arithmetic formulas.

1. Input employee details

2. Read basic salary

3.Calculate Allowances:

- HRA=20% of Basic Salary
- DA=10% of Basic Salary
- Bonus=5%
- Pf=8% deduction of salary

4.Calculate Gross Salary

Gross Salary=Basic Salary+HRA+DA+Bonus

5. Display Salary Details

4. Input and Output Handling

- cin for user input
- cout for displaying results

This allows interaction between user and the system

6. Problem Statement/Case study Implementation Details and Snapshots

Implementation Steps

1. Create a class called **Employee**.
2. Declare private variables:
- Employee ID
- Name
- Basic Salary
- Allowances
3. Create public member functions:
- inputDetails() – to enter employee data
- calculateSalary() – to compute salary components
- generatePayslip() – to display payslip
4. Implement a **menu-drive**

EmployeePayroll.cpp > main()

```cpp
#include <iostream>
#include <string>
using namespace std;
class Employee
{
private:
    int empID;
    string name;
    double basicSalary;
    double hra, da, pf, bonus;
    double netSalary;
public:
   void inputDetails()
    {
        cout << "Enter Employee ID:\n ";
        cin >> empID;
        cin.ignore();
        cout << "Enter Employee Name: ";
        getline(cin, name);

        cout << "Enter Basic Salary: ";
        cin >> basicSalary;
        calculateSalary();
    }

    void calculateSalary()
    {
        hra = 0.20 * basicSalary;
        da = 0.10 * basicSalary;
        bonus = 0.05 * basicSalary;
        pf = 0.12 * basicSalary;
        netSalary = basicSalary + hra + da + bonus - pf;
    }
     int getID()
    {
        return empID;
    }
     void generatePayslip()
    {
        cout << "\n========== Employee Payslip ==========\n";
```

```
        cout << "Employee ID : " << empID << endl;
        cout << "Name        : " << name << endl;
        cout << "----------------------------------------\n";
        cout << "Basic Salary : " << basicSalary << endl;
        cout << "HRA (20%)    : " << hra << endl;
        cout << "DA (10%)     : " << da << endl;
        cout << "Bonus (5%)   : " << bonus << endl;
        cout << "Gross Salary : " << grossSalary << endl;
        cout << "----------------------------------------\n";
        cout << "Net Salary   : " << netSalary << endl;
        cout << "========================================\n";
        cout << "Payslip Generated Successfully!\n";
    }
};
int main()
{
    Employee emp[50];
    int count = 0;
    int choice;
    int searchID;
    bool found;
    do
    {
        cout << "\n===== Employee Payroll System =====\n";
        cout << "1. Add Employee\n";
        cout << "2. Generate Payslip\n";
        cout << "3. Exit\n";
        cout << "Enter your choice: ";
        cin >> choice;
       switch(choice)
        {
            case 1:
                if(count < 50)
                {
                    emp[count].inputDetails();
                    count++;
                    cout << "\nEmployee Added Successfully\n";
                }
```

std::__1::ostream std::__1::cout

```
            else
            {
                cout << "\nEmployee limit reached\n";
            }
            break;
        case 2:
            cout << "\nEnter Employee ID to generate payslip: ";
            cin >> searchID;
           found = false;
            for(int i = 0; i < count; i++)
            {
                if(emp[i].getID() == searchID)
                {
                    emp[i].generatePayslip();
                    found = true;
                    break;
                }
            }
             if(!found)
            {
                cout << "Employee ID not found!\n";
            }
            break;

        case 3:
            cout << "\nExiting Program...\n";
            break;
         default:
            cout << "\nInvalid choice! Please try again.\n";
        }

    } while(choice != 3);

    return 0;
}
```

OUTPUT

```
dharmitsathvara@DHARMITs-MacBook-Air C++ % cd "/Users/dharmitsathvara/C++/" && g++ EmployeePayroll.cpp -o EmployeePayroll && "/Users/dha
itsathvara/C++/"EmployeePayroll

==== Employee Payroll System ====
1. Add Employee
2. Generate Payslip
3. Exit
Enter your choice: 1
Enter Employee ID:
 101
Enter Employee Name: Krishna Patel
Enter Basic Salary: 57800

Employee Added Successfully
```

This is the part where we add the employee details like id, name and basic salary of the Employee.

```
===== Employee Payroll System =====
1. Add Employee
2. Generate Payslip
3. Exit
Enter your choice: 2

Enter Employee ID to generate payslip: 101

========== Employee Payslip ==========
Employee ID : 101
Name        : Krishna Patel
----------------------------------------
Basic Salary : 57800
HRA (20%)    : 11560
DA (10%)     : 5780
Bonus (5%)   : 2890
PF (12%)     : -6936
----------------------------------------
Net Salary   : 71094
========================================
Payslip Generated Successfully!
```

Here the Calculation part works and after adding the Emp id it gives the total net salary after all allowances and deductions.

7. Problem Statement / Case Study Results and Conclusion

The **Employee Payroll System** successfully demonstrates how programming can be used to automate salary calculations and manage employee payroll data.

The program allows users to:

4. Enter employee details
5. Calculate salary automatically
6. Generate a formatted payslip

Through this case study, the following concepts were effectively applied:

- Object-Oriented Programming (Classes and Objects)
- Function-based program structure
- Menu-driven program design
- Basic arithmetic computations

The system provides a simple yet effective demonstration of payroll management logic. It also highlights how programming can reduce manual errors and improve efficiency in administrative tasks.

Future improvements could include:

- Managing multiple employees
- Adding tax deductions
- Saving payroll data to files or databases
- Creating a graphical user interface

Overall, the case study provides practical experience in applying **C++ programming concepts to solve real-world problems**.

References

1. C++ Programming Language Documentation
2.W3Schools.com
3.Microsoft C++ Learn documentation